

Supplementary materials for “Revealing the Interstitial-Mediated Sluggish Diffusion Mechanism in Concentrated Solid-Solution Alloys via Machine Learning-Integrated Kinetic Monte Carlo”

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Supplementary Note 1: The invariance properties

Permutation Invariance

In atomic systems, permutation invariance means that the model's prediction remains unchanged if atoms of the same type are permuted. In our approach, the input is a sorted sequence of atom types. If two atoms of the same type swap their positions in this sequence, the corresponding encoding remains identical, thus preserving permutation invariance. For instance, in a binary Fe–Ni system, any Fe atom in the sequence can be encoded as “0,” so exchanging two Fe atoms does not alter the encoded information and therefore does not affect the predicted energy barriers.

Translation Invariance

Translation invariance implies that shifting the entire atomic system in space does not alter the predicted energy barrier. This is ensured by how we sort atoms relative to the position of the interstitial (i.e., the “gap” or defect). The distance information used by the ICST method is based on a coordinate system defined by the interstitial site. Because all atoms are referenced to the same interstitial-centered coordinate system, any global shift of the system leaves the relative positions unchanged.

Rotation Invariance

Finally, rotation invariance means that rotating the entire atomic configuration does not affect the predicted energy barriers. In our approach, the interstitial atom's relative coordinates before and after migration remain consistent in the interstitial-based coordinate system. Hence, rotating the system does not change the atoms' positions in this relative coordinate frame, preserving rotational invariance.

Supplementary Note 2: Prefactor determination in Fe-Ni CSAs

Based on the literature referenced by the reviewer and related studies[1-3] , it is clear that in KMC simulations of multi-principal element alloys (MPEAs), the prefactor (or attempt frequency) is significantly influenced by the diverse configurations' energy and vibrational entropy, which can strongly affect the resulting diffusion coefficients.

In our ML-KMC simulations, the determination of the prefactor or attempt frequency follows the method described in [4]. Specifically, we compared the diffusion coefficients ($D_{\text{ML-KMC}}$) obtained from ML-KMC simulations at a constant attempt frequency (AF) of 10 THz at 1000 K with the diffusion coefficients derived from long-time MD simulations (D_{MD}) at the same temperature. The ratio $D_{\text{ML-KMC}}/D_{\text{MD}}$ was fitted as a high-ord polynomial $f(x_{\text{Fe}})$ where x_{Fe} represents the Fe composition. This fitting function represents the prefactor's impact on the diffusion coefficients between the MD and constant AF KMC at 1000K. The

corresponding attempt frequency values for different compositions, based on the $D_{\text{ML-KMC}}/D_{\text{MD}}$ ratio, are presented in **Table 1**.

Table 1 The prefactors (attempt frequencies) determined by the $D_{\text{ML-KMC}}/D_{\text{MD}}$ with different Fe components of Fe-Ni alloy in the ML-kMC method

Fe component	Attempt frequency(THz)	Fe component	Attempt frequency(THz)
0	3.1426	0.550	4.6173
0.025	3.1317	0.600	4.5855
0.050	3.2123	0.650	5.8204
0.100	3.1315	0.700	7.2354
0.200	3.5033	0.750	10.0724
0.250	4.4667	0.800	12.3312
0.350	9.4584	0.900	18.0203
0.450	6.8101	0.950	17.0439
0.500	5.2636	1	15.9040

Furthermore, based on [4], the attempt frequency at different temperatures T can be determined using the relationship: $\nu_{0,T} = \frac{T}{1000} \nu_{0,1000}$. It is important to note that when applying ML-KMC to predict diffusion properties in other multi-principal element systems, this method does not directly account for the influence of the prefactor. Therefore, it is essential to use the ratio of diffusion coefficients ($D_{\text{ML-KMC}}/D_{\text{MD}}$) obtained from ML-KMC simulations at high temperatures with a constant attempt frequency and MD diffusion simulations to incorporate the effect of the prefactor.

Supplementary Note 3: The energy barrier influences diffusion behavior in Fe-Ni CSAs

All the complex variation behaviour with the crucial diffusion factors in **Figure 4** is mainly sourced from the energy barriers, which can be categorized into eight types of migration energy distributions according to the element types before and after the migration of dumbbell interstitials, as shown in **Figure 6**. Note that the eight migration modes of dumbbell interstitials are not independent, and their mutual transformation relationships are shown in **Figure 5a**. For instance, the “Ni-Ni” interstitial dumbbell can migrate to form

“Ni-Ni” or “Ni-Fe” but not “Fe-Ni” or “Fe-Fe”. In this figure., the double-sided arrow indicates that they can mutually transform (e.g., “Ni-Ni-Ni” or “Ni-Fe-Ni”), while the one-way arrow means they can only transform in a single direction (e.g., “Ni-Ni-Fe” or “Ni-Fe-Fe”). The arrows with the same colour represent the same migration events. The migration probability of the different migration modes under different components is introduced to help the analysis of the complex behaviors, as shown in **Figure 5b**. The probability is calculated by the share of the number of specific migration events in the total migration steps. For example, the probability of “Ni-Ni-Fe” corresponds to the fraction of the event where the initial migrating interstitial dumbbell “Ni-Ni” transforms to the “Ni-Fe”. With the above information, the detailed explanation of the variation trend of the tracer correlation factor (f_{tr}) and jump frequency (ν) is elaborated as follows.

(1) Tracer correlation factor

Figure. 4 in the manuscript shows the change of the f_{tr} , which firstly decreases in the composition range $x \in [0, 0.6]$ and then increases for $x \in [0.6, 1]$. From **Figure 5b** in the manuscript, we found that the probability of “Ni-Ni-Ni” is near 1 in $Fe_{0.1}Ni_{0.9}$, suggesting that there is only “Ni-Ni-Ni” interstitial transition contributing to diffusion. Compared with pure Ni, the rising Fe component leads to an increase in the potential forward-jumping paths of “Ni-Ni-Fe” and simultaneously causes a decrease in the backward-jumping paths of “Ni-Ni-Ni”. Thus, the backward-jumping probability of “Ni-Ni-Ni” increases and strengthens the correlation effect, leading to the low correlation factor at such a component.

With the further increase of Fe components higher than 0.1, the high Fe components in the atomic environment enhance the “Ni-Ni-Fe” transitions, though “Ni-Ni-Ni” transitions are still dominant. **Figure 5a** in the manuscript shows that after the “Ni-Ni-Fe” transition happens, there are four possible transition paths: “Fe-Ni-Ni”, “Fe-Ni-Fe”, “Ni-Fe-Ni”, and “Ni-Fe-Fe”. Among them, only “Fe-Ni-Ni” would happen since the jump probability of other paths is almost 0, as shown in **Figure 5b** of the manuscript. In poor Fe components, such as $Fe_{0.1}Ni_{0.9}$, an average of 7.2 in 8 possible transition paths are “Fe-Ni-Ni”. With the Fe components rising, the average transition paths of “Fe-Ni-Ni” gradually decrease, which increases the backward-jumping probability. Thus, this behaviour contributes to a strengthening correlation effect. On the other hand, the increasing jump probability of “Fe-Ni-Fe” suggests that part of the “Fe-Ni-Ni” transition is gradually substituted after the “Ni-Ni-Fe” transition happens. The growing transition choice decreases the backward-jumping probability of “Fe-Ni-Ni”, leading to a weakening correlation effect. At $Fe_{0.6}Ni_{0.4}$, nearly half of the original “Fe-Ni-Ni” transitions after the Ni-Ni-Fe happens are replaced by the “Fe-Ni-Fe”, which suggests that the weakening and enhancing correlation effect may cancel each other out in these transitions. However, the enhancing correlation effect from the “Ni-Ni-Ni” transitions still works. Therefore, there is a gradually strengthening correlation effect in $x \in [0, 0.6]$, as **Figure 4** in the manuscript shows.

A transition in the correlation factor happens at $x=0.6$. This is because, at $\text{Fe}_{0.7}\text{Ni}_{0.3}$, there is an almost equal jump probability with the “Fe-Ni-Fe”, “Fe-Ni-Ni”, “Ni-Ni-Ni”, and “Ni-Ni-Fe”, which means there is less bias transition among these transitions. This behaviour contributes to a significant weakening in the correlation effect than $x=0.6$. Besides, “Fe-Fe-Fe” shows a much higher jump probability than “Ni-Fe-Fe” and “Fe-Fe-Ni” in this regime, which indicates that once the Ni-Ni-Fe to Ni-Fe-Fe jump happens, the interstitial dumbbell “Fe-Fe” prefers to form new “Fe-Fe” but not to produce “Fe-Ni” interstitial dumbbells. With the Fe component increase, there are rising “Fe-Fe-Fe” transition paths that lead to a reduction of backward jumping probability. Thus, a reverse trend in the correlation factor is found starting from $\text{Fe}_{0.6}\text{Ni}_{0.4}$.

(2) Jump frequency

In **Figure 4** of the manuscript, the jump frequency first increases in $x \in [0.1, 0.4]$, then decreases in $x \in [0.4, 0.6]$, and finally rises again in $x \in [0.6, 0.9]$. Therefore, we analyze these three regimes separately.

The complex variation behaviour of jump frequency is mainly affected by the energy barrier distributions based on different interstitial transitions in **Figure 6** of the manuscript. Note that the area of the barrier distributions of each transition can reflect the probability of each migrating event around the migrating interstitial during KMC. For example, suppose the distribution mainly consists of the area of “Ni-Ni-Ni” and “Ni-Ni-Fe”, where the previous is several folds than the latter. In that case, the probability of “Ni-Ni-Ni” migrating paths around the “Ni-Ni” dumbbell is several folds than the “Ni-Ni-Fe” during the KMC. Thus, a comparison of the area dictates atomic environment information about how many possible migrating paths around the migrating dumbbell during the KMC process. In addition, the left-shifted energy distributions suggest smaller energy barriers, which contribute to the increase in the jump frequency according to Eq. (4) and (5). Thus, the changing area and the shifting energy distributions of each migrating event determine the variation trend of the jump frequency.

From **Figure. 6a to 6d** of the manuscript, “Ni-Ni-Ni” and “Ni-Ni-Fe” have a much larger area than other interstitial transitions. Therefore, the jump frequency is mainly dominated by these two transition events in $x \in [0.1, 0.4]$. These two distributions both have a left shift with the increasing Fe components, contributing to the increase in the jump frequency in this regime. Besides, the portion of “Fe-Ni-Ni”, located at even lower energies, also increases, which is consistent with the increase in the jump frequency. However, the decreasing count area of low-energy “Ni-Ni-Ni” is mainly replaced by the increasing area of high-energy “Ni-Ni-Fe”, resulting in the slower increase of the jump frequency in $x \in [0.1, 0.4]$.

In $x \in [0.4, 0.6]$, the area of high-energy “Ni-Ni-Fe” continuously increases and exhibits the largest area than other interstitial transitions, playing the dominant role in these components. Thus, although there is also an

increased area of low-energy “Fe-Ni-Ni” and “Fe-Fe-Ni”, the jump frequency shows a decreasing trend in this regime.

When the Fe components increase from 0.6 to 0.7, the “Ni-Ni-Fe” count area decreases. The reducing count area is replaced by the exponentially increasing “Fe-Fe-Fe” and “Fe-Ni-Fe” fractions, which are in a relatively lower energy region. These changes contribute to the significant increase in the jump frequency in these components. Besides, when $x \in [0.7, 0.9]$, the count area of “Fe-Fe-Fe” replaces the one of relatively higher energy “Ni-Ni-Ni” and “Ni-Ni-Fe”, and becomes dominant, leading to a continual increase of the jump frequency.

Therefore, the change in jump frequency can be understood by the corresponding energy barrier distributions. The turning point at $x=0.6$ is due to the exponential decrease of the higher-energy Ni-Ni-Ni(Fe) transitions mainly substituted by the lower-energy Fe-Fe(Ni)-Fe, leading to sluggish diffusion together with the lowest correlation factor at this composition.

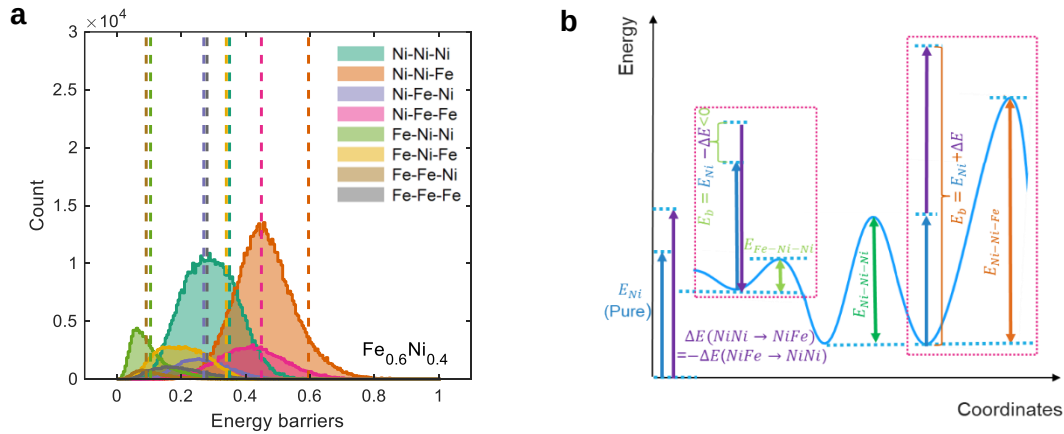
Supplementary Note 4: Energy barrier influence on migration ratio

The migration ratio of “Ni-Ni-Fe” always matches well with the “Fe-Ni-Ni”, because the “Ni-Ni-Fe” and “Fe-Ni-Ni” is the only mutual transition way between the “Ni-Ni” dumbbell to “Ni-Fe”. Although the “Ni-Fe” dumbbell generated by the “Ni-Ni-Fe” migration pattern can transfer to other types of dumbbells such as “Ni-Fe” or “Fe-Fe”, it always transfers back through the “Fe-Ni-Ni” migration pattern. For example, in **Figure 5b** of the manuscript at $Fe_{0.6}Ni_{0.4}$, the migration ratio of “Ni-Ni-Fe” almost equals to the “Fe-Ni-Ni”, and there is an obvious migration ratio with the “Fe-Ni-Fe”. To explain this phenomenon, we need to analyze the energy barrier information of different migration patterns. From **Figure 6f** of the manuscript, the migration energy barrier distribution of “Ni-Ni-Fe” has a large area and is located at the rightmost part of the figure, which suggests that there are many optional “Ni-Ni-Fe” transition paths around the “Ni-Ni” dumbbell, the higher energy barrier lead to the happening of “Ni-Ni-Fe” is rare. Even this rare “Ni-Ni-Fe” happens, the newly formed “Ni-Fe” dumbbell has four possible transition ways, “Fe-Ni-Fe”, “Fe-Ni-Ni”, “Ni-Ni-Ni”, and “Ni-Ni-Fe”. Among them, the “Fe-Ni-Ni” is located at the leftmost part of the figure, then the much lower energy barrier drives the “Ni-Fe” dumbbell transfer back to the “Ni-Ni” dumbbell. However, the “Fe-Ni-Fe” transition has a larger and larger overlap area with the “Fe-Ni-Ni”, which suggests there is a small possibility that the “Fe-Ni-Fe” transition happens.

Supplementary Note 5: Difference between ML-KMC and Ref-KMC

Although both ML-KMC and the simplified Ref-KMC reproduce the trend of compositional-dependent diffusion coefficients, they still show a significant difference. This difference between these two methods is mainly from the effective energy barrier information entered into the KMC process. In order to explain this difference clearly, we compare these two methods in detail.

Since the dumbbell types have a significant influence on energy barriers, it is necessary to consider the energy distributions for different dumbbell transitions. **Fig. 1a** and **1b** show the energy barrier distributions obtained from ML-KMC based on the eight different dumbbell transitions within representative alloy compositions with the $\text{Fe}_{0.6}\text{Ni}_{0.4}$ case. Besides, the effective E_b from the Ref-KMC approach is also illustrated by the dotted lines; the same color represents the same interstitial transition. For example, the pink dotted line from Ref-KMC in **Fig. 1a** corresponds to the pink “Ni-Fe-Fe” energy barrier distributions from ML-KMC. The Ref-KMC energy barrier is determined by the Bell-Evans-Polanyi principle. In this principle, if the dumbbell type is the same, then the migration energy is approximately equal to the jump barrier of the pure element type, for instance, Fe-Ni dumbbell to Fe-Ni dumbbell, Ni atom jumps, here can be named as Fe-Ni-Fe migration event, the migration energy is same as the one in pure Ni (E_{Ni}). On the other hand, if the dumbbell type before and after the jump is different, like Ni-Ni-Fe case, the migration energy $E_{\text{Ni-Ni-Fe}}$ is determined by the E_{Ni} and the difference of average dumbbell formation energy (ΔE) between Ni-Ni and Ni-Fe.



Figs. 1 Comparison of the energy barrier (E_b) distribution from ML-kMC and Ref-KMC. (a) $\text{Fe}_{0.6}\text{Ni}_{0.4}$ (b) The schematic diagram of the energy change in the transition process from “Fe-Ni-Ni” to “Ni-Ni-Fe”, and to “Ni-Ni-Fe” in ML-kMC ($E_{\text{Fe-Ni-Ni}}$, $E_{\text{Ni-Ni-Fe}}$, and $E_{\text{Ni-Ni-Fe}}$) and Ref-KMC.

As **Figs. 1a** show, the average energy barrier (dot lines) from the Ref-KMC and the average energy barriers of the distributions (approximately Gaussian distribution) from the ML-KMC have a consistent trend with

each other, for example, the $E_{\text{Fe-Ni-Ni}}$, $E_{\text{Fe-Fe-Ni}}$ in both the ML-KMC and Ref-KMC has a small energy barrier while the $E_{\text{Ni-Fe-Fe}}$, $E_{\text{Ni-Ni-Fe}}$ have relative larger barriers among different energy barriers. This consistency may lead to a similar trend in diffusivity predictions.

However, there is a significant discrepancy between these two methods. On the one hand, in Ref-KMC, the energy barrier of the “Ni-Ni-Ni” or “Fe-Ni-Fe” and “Fe-Fe-Fe” or “Ni-Fe-Ni” are assumed to have the same energy barrier as in the pure Ni (3.5 eV) and Fe (2.7 eV) [5, 6], respectively. In contrast, in ML-KMC, the energy barrier distributions of “Ni-Ni-Ni” or “Fe-Ni-Fe” and “Fe-Fe-Fe” or “Ni-Fe-Ni” exhibit large variations and show significant differences from those in pure Ni or Fe.

On the other hand, when the newly formed interstitial dumbbell is different from the initial interstitial dumbbell, e.g., Ni-Ni-Fe or Fe-Fe-Ni, the energy difference ΔE is added or subtracted to $E_{\text{Ni/Fe}}$ as the transition energy barrier ($E_{\text{Ni/Fe}} \pm \Delta E$) in Ref-KMC, these energy barriers information may not be accurate. Since ΔE decreases with the increasing Fe components [5], the sum ($E_{\text{Ni}} - \Delta E$) may cause a negative transition energy barrier in Ni-rich components, which is unreasonable. For example, the light green dotted line “Fe-Ni-Ni” in Ref-KMC is an artificially modified value (if $E_b < 0$, $E_b = 1 \times 10^{-3}$ eV) at $\text{Fe}_{0.4}\text{Ni}_{0.6}$ according to the reference [5]. To vividly illustrate this point, we show in **Figs. 1b** the schematic energy diagram for the transition of “Fe-Ni-Ni”. For this transition, the energy difference from Ni-Fe to Ni-Ni dumbbell is a large negative value (ΔE), leading to a negative transition energy barrier ($E_b = E_{\text{Ni}} - \Delta E < 0$) in Ref-KMC, which is in stark contrast to the energy barrier distributions ($E_{\text{Fe-Ni-Ni}}$) in ML-KMC. Such simplification also causes a much higher transition energy barrier ($E_{\text{Ni}} + \Delta E$) for the reverse migrating process “Ni-Ni-Fe”, as shown in the right part of **Figs. 1b**.

Reference

- [1] J. Lefèvre López, N. Mousseau, G. Adjanor, and C. Domain, "Harmonic transition state theory applied to vacancy diffusion pre-exponential factors in a concentrated solid-solution alloy," *Physical Review Materials*, vol. 8, no. 1, p. 013609, 01/30/ 2024, doi: 10.1103/PhysRevMaterials.8.013609.
- [2] S. Gelin, A. Champagne-Ruel, and N. Mousseau, "Enthalpy-entropy compensation of atomic diffusion originates from softening of low frequency phonons," *Nature Communications*, vol. 11, no. 1, p. 3977, 2020/08/07 2020, doi: 10.1038/s41467-020-17812-2.
- [3] A. Sauvé-Lacoursière, S. Gelin, G. Adjanor, C. Domain, and N. Mousseau, "Unexpected role of prefactors in defects diffusion: The case of vacancies in the 55Fe-28Ni-17Cr concentrated solid-solution alloys," *Acta Materialia*, vol. 237, p. 118153, 2022/09/15/ 2022, doi: <https://doi.org/10.1016/j.actamat.2022.118153>.
- [4] W. Huang and X.-M. Bai, "Machine Learning Based On-The-Fly Kinetic Monte Carlo Simulations of Sluggish Diffusion in Ni-Fe Concentrated Alloys," *Journal of Alloys and Compounds*, p. 168457, 2022.
- [5] K. Ferasat, Y. N. Osetsky, A. V. Barashev, Y. Zhang, Z. Yao, and L. K. Béland, "Accelerated kinetic Monte Carlo: A case study; vacancy and dumbbell interstitial diffusion traps in concentrated solid solution alloys," *The Journal of Chemical Physics*, vol. 153, no. 7, p. 074109, 2020.
- [6] Y. Osetsky, A. V. Barashev, L. K. Béland, Z. Yao, K. Ferasat, and Y. Zhang, "Tunable chemical complexity to control atomic diffusion in alloys," *npj Computational Materials*, vol. 6, no. 1, p. 38, 2020/04/21 2020, doi: 10.1038/s41524-020-0306-9.